

## ANEXO I

## A2 CORREÇÃO DE ALTURA DE 10° - 90° — SOL, ESTRELAS E PLANETAS

| SOL       |             |           |             | ESTRELAS E PLANETAS |       |                 |           | DEPRESSÃO |       |         |               |
|-----------|-------------|-----------|-------------|---------------------|-------|-----------------|-----------|-----------|-------|---------|---------------|
| Out — Mar |             | Abr — Set |             |                     |       |                 |           |           |       |         |               |
| a         | Limbo       | a         | Limbo       | a                   | Corr. | a               | Corr.     | Elev do   | Corr. | Elev do | Elev do       |
| ap        | Inf Sup     | ap        | Inf Sup     | ap                  |       | ap              | adicional | Olho      |       | Olho    | Olho          |
|           |             |           |             |                     |       |                 |           |           |       |         |               |
| 9 33      | +10.8 -21.5 | 9 39      | +10.6 -21.2 | 9 55                |       | <b>2011</b>     |           | m         |       | Pés     | m             |
| 9 45      | +10.9 -21.4 | 9 50      | +10.7 -21.1 | 10 07               | -5.3  | <b>VÊNUS</b>    |           | 2.4       | -2.8  | 8.6     | 1.0 - 1.8     |
| 9 56      | +11.0 -21.3 | 10 02     | +10.8 -21.0 | 10 20               | -5.2  | 1 Jan - 18 Fev  |           | 2.6       | -2.9  | 8.6     | 1.5 - 2.2     |
| 10 08     | +11.1 -21.2 | 10 14     | +10.9 -20.9 | 10 32               | -5.1  | 0               |           | 2.8       | -3.0  | 9.2     | 2.0 - 2.5     |
| 10 20     | +11.2 -21.1 | 10 27     | +11.0 -20.8 | 10 46               | -5.0  | 0 +0.2          |           | 3.0       | -3.1  | 9.8     | 2.5 - 2.8     |
| 10 33     | +11.3 -21.0 | 10 40     | +11.1 -20.7 | 10 59               | -4.9  | 41 +0.1         |           | 3.2       | -3.2  | 10.5    | 3.0 - 3.0     |
| 10 46     | +11.4 -20.9 | 10 53     | +11.2 -20.6 | 11 14               | -4.8  | 76 +0.1         |           | 3.4       | -3.3  | 11.2    | Ver tábu<br>← |
| 11 00     | +11.5 -20.8 | 11 07     | +11.3 -20.5 | 11 29               | -4.7  | 19 Fev - 31 Dez |           | 3.6       | -3.4  | 11.9    |               |
| 11 15     | +11.6 -20.7 | 11 22     | +11.4 -20.4 | 11 44               | -4.6  | 0               |           | 3.8       | -3.5  | 12.6    | m             |
| 11 30     | +11.7 -20.6 | 11 37     | +11.5 -20.3 | 12 00               | -4.5  | 0 +0.1          |           | 4.0       | -3.6  | 13.3    |               |
| 11 45     | +11.8 -20.5 | 12 10     | +11.6 -20.2 | 12 17               | -4.4  | <b>MARTE</b>    |           | 4.3       | -3.7  | 14.1    | 20 - 7.9      |
| 12 01     | +11.9 -20.4 | 12 27     | +11.7 -20.1 | 12 35               | -4.3  | 1 Jan - 31 Dez  |           | 4.5       | -3.8  | 14.9    | 22 - 8.3      |
| 12 18     | +12.0 -20.3 | 12 45     | +11.8 -20.0 | 12 53               | -4.2  | 0               |           | 4.7       | -3.9  | 15.7    | 24 - 8.6      |
| 12 36     | +12.1 -20.2 | 13 04     | +11.9 -19.9 | 13 12               | -4.1  | 0 +0.1          |           | 5.0       | -4.0  | 16.5    | 26 - 9.0      |
| 12 54     | +12.2 -20.1 | 13 24     | +12.0 -19.8 | 13 32               | -4.0  | 60              |           | 5.2       | -4.1  | 17.4    | 28 - 9.3      |
| 13 14     | +12.3 -20.0 | 13 44     | +12.1 -19.7 | 13 53               | -3.9  |                 |           | 5.5       | -4.2  | 18.3    | 30 - 9.6      |
| 13 34     | +12.4 -19.9 | 14 06     | +12.2 -19.6 | 14 16               | -3.8  |                 |           | 5.8       | -4.3  | 19.1    | 32 - 10.0     |
| 13 55     | +12.5 -19.8 | 14 29     | +12.3 -19.5 | 14 39               | -3.7  |                 |           | 6.1       | -4.4  | 20.1    | 34 - 10.3     |
| 14 17     | +12.6 -19.7 | 14 53     | +12.4 -19.4 | 15 03               | -3.6  |                 |           | 6.3       | -4.5  | 21.0    | 36 - 10.6     |
| 14 41     | +12.7 -19.6 | 15 18     | +12.5 -19.3 | 15 29               | -3.5  |                 |           | 6.6       | -4.6  | 22.0    | 38 - 10.8     |
| 15 05     | +12.8 -19.5 | 15 45     | +12.6 -19.2 | 15 56               | -3.4  |                 |           | 6.9       | -4.7  | 22.9    | Pés           |
| 15 31     | +12.9 -19.4 | 16 13     | +12.7 -19.1 | 16 25               | -3.3  |                 |           | 7.2       | -4.8  | 23.9    |               |
| 15 59     | +13.0 -19.3 | 16 43     | +12.8 -19.0 | 16 55               | -3.2  |                 |           | 7.5       | -4.9  | 24.9    | 40 - 11.1     |
| 16 27     | +13.1 -19.2 | 17 14     | +12.9 -18.9 | 17 27               | -3.1  |                 |           | 7.9       | -5.0  | 26.0    | 42 - 11.4     |
| 16 58     | +13.2 -19.1 | 17 47     | +13.0 -18.8 | 18 01               | -3.0  |                 |           | 8.2       | -5.1  | 27.1    | 44 - 11.7     |
| 17 30     | +13.3 -19.0 | 18 23     | +13.1 -18.7 | 18 37               | -2.9  |                 |           | 8.5       | -5.2  | 28.1    | 46 - 11.9     |
| 18 05     | +13.4 -18.9 | 19 00     | +13.2 -18.6 | 19 16               | -2.8  |                 |           | 8.8       | -5.3  | 29.2    | 48 - 12.2     |
| 18 41     | +13.5 -18.8 | 19 41     | +13.3 -18.5 | 19 56               | -2.7  |                 |           | 9.2       | -5.4  | 30.4    | Ver tábu<br>← |
| 19 20     | +13.6 -18.7 | 20 24     | +13.4 -18.4 | 20 40               | -2.6  |                 |           | 9.5       | -5.5  | 31.5    |               |
| 20 02     | +13.7 -18.6 | 21 10     | +13.5 -18.3 | 21 27               | -2.5  |                 |           | 9.9       | -5.6  | 32.7    | ft.           |
| 20 46     | +13.8 -18.5 | 21 59     | +13.6 -18.2 | 22 17               | -2.4  |                 |           | 10.3      | -5.7  | 33.9    | 70 - 8.1      |
| 21 34     | +13.9 -18.4 | 22 52     | +13.7 -18.1 | 23 11               | -2.3  |                 |           | 10.6      | -5.8  | 35.1    | 75 - 8.4      |
| 22 25     | +14.0 -18.3 | 23 49     | +13.8 -18.0 | 24 09               | -2.2  |                 |           | 11.0      | -5.9  | 36.3    | 80 - 8.7      |
| 23 20     | +14.1 -18.2 | 24 51     | +13.9 -17.9 | 25 12               | -2.1  |                 |           | 11.4      | -6.0  | 37.6    | 85 - 8.9      |
| 24 20     | +14.2 -18.1 | 25 58     | +14.0 -17.8 | 26 20               | -2.0  |                 |           | 11.8      | -6.1  | 38.9    | 90 - 9.2      |
| 25 24     | +14.3 -18.0 | 27 11     | +14.1 -17.7 | 27 34               | -1.9  |                 |           | 12.2      | -6.2  | 40.1    | 95 - 9.5      |
| 26 34     | +14.4 -17.9 | 28 31     | +14.2 -17.6 | 28 54               | -1.8  |                 |           | 12.6      | -6.3  | 41.5    | Ver tábu<br>← |
| 27 50     | +14.5 -17.8 | 29 58     | +14.3 -17.5 | 30 22               | -1.7  |                 |           | 13.0      | -6.4  | 42.8    |               |
| 29 13     | +14.6 -17.7 | 31 33     | +14.4 -17.4 | 31 58               | -1.6  |                 |           | 13.4      | -6.5  | 44.2    | 100 - 9.7     |
| 30 44     | +14.7 -17.6 | 33 18     | +14.5 -17.3 | 33 43               | -1.5  |                 |           | 13.8      | -6.6  | 45.5    | 105 - 9.9     |
| 32 24     | +14.8 -17.5 | 35 15     | +14.6 -17.2 | 35 38               | -1.4  |                 |           | 14.2      | -6.7  | 46.9    | 110 - 10.2    |
| 34 15     | +14.9 -17.4 | 37 24     | +14.7 -17.1 | 37 45               | -1.3  |                 |           | 14.7      | -6.8  | 48.4    | 115 - 10.4    |
| 36 17     | +15.0 -17.3 | 39 48     | +14.8 -17.0 | 40 06               | -1.2  |                 |           | 15.1      | -6.9  | 49.8    | 120 - 10.6    |
| 38 34     | +15.1 -17.2 | 42 28     | +14.9 -16.9 | 42 42               | -1.1  |                 |           | 15.5      | -7.0  | 51.3    | 125 - 10.8    |
| 41 06     | +15.2 -17.1 | 45 29     | +15.0 -16.8 | 45 34               | -1.0  |                 |           | 16.0      | -7.1  | 52.8    | 130 - 11.1    |
| 43 56     | +15.3 -17.0 | 48 52     | +15.1 -16.7 | 48 45               | -0.9  |                 |           | 16.5      | -7.2  | 54.3    |               |
| 47 07     | +15.4 -16.9 | 52 41     | +15.2 -16.6 | 52 16               | -0.8  |                 |           | 16.9      | -7.3  | 55.8    | 135 - 11.3    |
| 50 43     | +15.5 -16.8 | 56 59     | +15.3 -16.5 | 56 09               | -0.7  |                 |           | 17.4      | -7.4  | 57.4    | 140 - 11.5    |
| 54 46     | +15.6 -16.7 | 61 50     | +15.4 -16.4 | 60 26               | -0.6  |                 |           | 17.9      | -7.5  | 58.9    | 145 - 11.7    |
| 59 21     | +15.7 -16.6 | 67 15     | +15.5 -16.3 | 65 06               | -0.5  |                 |           | 18.4      | -7.6  | 60.5    | 150 - 11.9    |
| 64 28     | +15.8 -16.5 | 73 14     | +15.6 -16.2 | 70 09               | -0.4  |                 |           | 18.8      | -7.7  | 62.1    | 155 - 12.1    |
| 70 10     | +15.9 -16.4 | 79 42     | +15.7 -16.1 | 75 32               | -0.3  |                 |           | 19.3      | -7.8  | 63.8    |               |
| 76 24     | +16.0 -16.3 | 86 31     | +15.8 -16.0 | 81 12               | -0.2  |                 |           | 19.8      | -7.9  | 65.4    |               |
| 83 05     | +16.1 -16.2 | 90 00     | +15.9 -15.9 | 87 03               | -0.1  |                 |           | 20.4      | -8.0  | 67.1    |               |
| 90 00     |             |           |             | 90 00               | 0.0   |                 |           | 20.9      | -8.1  | 68.8    |               |
|           |             |           |             |                     |       |                 |           | 21.4      |       | 70.5    |               |

a ap = Altura dada pelo sextante corrigida do erro instrumental e da depressão



## ANEXO II

5, 6 e 7 DE AGOSTO DE 2011 (6ª feira, Sábado e Domingo)

157

| TU                                         | SOL   |                       | LUA                    |               |                   |       |       | Lat   | CREP  |       | SOL<br>Nascer | LUA – Nascer |     |     |     |
|--------------------------------------------|-------|-----------------------|------------------------|---------------|-------------------|-------|-------|-------|-------|-------|---------------|--------------|-----|-----|-----|
|                                            | AHG   | Dec                   | AHG                    | v             | Dec               | d     | Ph    |       | Naut  | Civil |               | 5            | 6   | 7   | 8   |
| d h                                        | °     | '                     | °                      | '             | °                 | '     |       | h m   | h m   | h m   | h m           | h m          | h m | h m | h m |
| SEX<br>T<br>A<br><br>F<br>E<br>I<br>R<br>A | N 72  | ////                  | ////                   |               |                   |       |       | 00 51 | 15 05 | ██    | ██            | ██           |     |     |     |
|                                            | N 70  | ////                  | ////                   |               |                   |       |       | 02 02 | 14 21 | ██    | ██            | ██           |     |     |     |
|                                            | 68    | ////                  | ////                   |               |                   |       |       | 02 38 | 13 52 | 16 05 | ██            | ██           |     |     |     |
|                                            | 66    | ////                  |                        | 01 20         | 03 03             | 13 31 | 15 26 | 17 20 | 18 54 |       |               |              |     |     |     |
|                                            | 64    | ////                  |                        | 02 05         | 03 23             | 13 14 | 14 59 | 16 38 | 17 57 |       |               |              |     |     |     |
|                                            | 62    | ////                  |                        | 02 33         | 03 38             | 13 00 | 14 38 | 16 09 | 17 24 |       |               |              |     |     |     |
|                                            | 60    | 01 12                 | 02 54                  | 03 52         | 12 48             | 14 22 | 15 48 | 16 59 |       |       |               |              |     |     |     |
|                                            | N 58  | 01 53                 | 03 12                  | 04 03         | 12 38             | 14 07 | 15 30 | 16 39 |       |       |               |              |     |     |     |
|                                            | 56    | 02 19                 | 03 26                  | 04 13         | 12 30             | 13 55 | 15 15 | 16 23 |       |       |               |              |     |     |     |
|                                            | 54    | 02 39                 | 03 38                  | 04 21         | 12 22             | 13 45 | 15 02 | 16 09 |       |       |               |              |     |     |     |
|                                            | 52    | 02 55                 | 03 49                  | 04 29         | 12 15             | 13 36 | 14 51 | 15 57 |       |       |               |              |     |     |     |
|                                            | 50    | 03 09                 | 03 58                  | 04 36         | 12 09             | 13 27 | 14 41 | 15 46 |       |       |               |              |     |     |     |
|                                            | 45    | 03 36                 | 04 17                  | 04 51         | 11 55             | 13 10 | 14 20 | 15 24 |       |       |               |              |     |     |     |
|                                            | N 40  | 03 56                 | 04 33                  | 05 03         | 11 44             | 12 55 | 14 03 | 15 06 |       |       |               |              |     |     |     |
|                                            | 35    | 04 12                 | 04 45                  | 05 13         | 11 35             | 12 43 | 13 49 | 14 51 |       |       |               |              |     |     |     |
|                                            | 30    | 04 25                 | 04 56                  | 05 22         | 11 27             | 12 32 | 13 37 | 14 38 |       |       |               |              |     |     |     |
|                                            | 20    | 04 46                 | 05 14                  | 05 37         | 11 13             | 12 14 | 13 16 | 14 15 |       |       |               |              |     |     |     |
|                                            | N 10  | 05 02                 | 05 28                  | 05 50         | 11 00             | 11 59 | 12 58 | 13 56 |       |       |               |              |     |     |     |
|                                            | 0     | 05 16                 | 05 41                  | 06 02         | 10 49             | 11 44 | 12 41 | 13 38 |       |       |               |              |     |     |     |
|                                            | S 10  | 05 27                 | 05 53                  | 06 15         | 10 38             | 11 30 | 12 24 | 13 20 |       |       |               |              |     |     |     |
|                                            | 20    | 05 38                 | 06 04                  | 06 27         | 10 26             | 11 14 | 12 06 | 13 01 |       |       |               |              |     |     |     |
|                                            | 30    | 05 48                 | 06 17                  | 06 42         | 10 12             | 10 57 | 11 46 | 12 39 |       |       |               |              |     |     |     |
|                                            | 35    | 05 54                 | 06 24                  | 06 50         | 10 05             | 10 47 | 11 34 | 12 26 |       |       |               |              |     |     |     |
|                                            | 40    | 05 59                 | 06 31                  | 07 00         | 09 56             | 10 35 | 11 20 | 12 11 |       |       |               |              |     |     |     |
|                                            | 45    | 06 05                 | 06 40                  | 07 11         | 09 46             | 10 21 | 11 04 | 11 53 |       |       |               |              |     |     |     |
| S 50                                       | 06 11 | 06 50                 | 07 24                  | 09 33         | 10 05             | 10 44 | 11 32 |       |       |       |               |              |     |     |     |
| 52                                         | 06 14 | 06 54                 | 07 31                  | 09 28         | 09 57             | 10 34 | 11 21 |       |       |       |               |              |     |     |     |
| 54                                         | 06 16 | 06 59                 | 07 37                  | 09 21         | 09 49             | 10 24 | 11 09 |       |       |       |               |              |     |     |     |
| 56                                         | 06 19 | 07 04                 | 07 45                  | 09 14         | 09 39             | 10 12 | 10 56 |       |       |       |               |              |     |     |     |
| 58                                         | 06 23 | 07 10                 | 07 54                  | 09 06         | 09 28             | 09 58 | 10 40 |       |       |       |               |              |     |     |     |
| S 60                                       | 06 26 | 07 17                 | 08 04                  | 08 58         | 09 16             | 09 42 | 10 22 |       |       |       |               |              |     |     |     |
| S<br>Á<br>B<br>A<br>D<br>O                 | Lat   | SOL<br>Pôr            | CREP<br>Civil          | Naut          | LUA – Pôr         |       |       |       |       |       |               |              |     |     |     |
|                                            | 5     | 6                     | 7                      | 8             |                   |       |       |       |       |       |               |              |     |     |     |
| N 72                                       | 23 06 | ////                  | ////                   | 18 40         | ██                | ██    | ██    |       |       |       |               |              |     |     |     |
| N 70                                       | 22 05 | ////                  | ////                   | 19 25         | ██                | ██    | ██    |       |       |       |               |              |     |     |     |
| 68                                         | 21 30 | ////                  | ////                   | 19 55         | 19 38             | ██    | ██    |       |       |       |               |              |     |     |     |
| 66                                         | 21 06 | 22 44                 | ////                   | 20 17         | 20 18             | 20 23 | 20 49 |       |       |       |               |              |     |     |     |
| 64                                         | 20 47 | 22 03                 | ////                   | 20 35         | 20 45             | 21 05 | 21 46 |       |       |       |               |              |     |     |     |
| 62                                         | 20 32 | 21 36                 | ////                   | 20 50         | 21 07             | 21 34 | 22 19 |       |       |       |               |              |     |     |     |
| 60                                         | 20 19 | 21 15                 | 22 53                  | 21 03         | 21 24             | 21 56 | 22 44 |       |       |       |               |              |     |     |     |
| N 58                                       | 20 08 | 20 58                 | 22 16                  | 21 14         | 21 39             | 22 14 | 23 04 |       |       |       |               |              |     |     |     |
| 56                                         | 19 58 | 20 44                 | 21 50                  | 21 23         | 21 52             | 22 30 | 23 20 |       |       |       |               |              |     |     |     |
| 54                                         | 19 49 | 20 32                 | 21 31                  | 21 32         | 22 03             | 22 43 | 23 34 |       |       |       |               |              |     |     |     |
| 52                                         | 19 42 | 20 22                 | 21 15                  | 21 39         | 22 12             | 22 54 | 23 46 |       |       |       |               |              |     |     |     |
| 50                                         | 19 35 | 20 13                 | 21 01                  | 21 46         | 22 21             | 23 04 | 23 57 |       |       |       |               |              |     |     |     |
| 45                                         | 19 21 | 19 54                 | 20 35                  | 22 01         | 22 40             | 23 26 | 24 19 |       |       |       |               |              |     |     |     |
| N 40                                       | 19 09 | 19 38                 | 20 15                  | 22 13         | 22 55             | 23 43 | 24 37 |       |       |       |               |              |     |     |     |
| 35                                         | 18 59 | 19 26                 | 19 59                  | 22 23         | 23 08             | 23 57 | 24 53 |       |       |       |               |              |     |     |     |
| 30                                         | 18 50 | 19 15                 | 19 46                  | 22 33         | 23 19             | 24 10 | 00 10 |       |       |       |               |              |     |     |     |
| 20                                         | 18 35 | 18 58                 | 19 25                  | 22 48         | 23 38             | 24 32 | 00 32 |       |       |       |               |              |     |     |     |
| N 10                                       | 18 22 | 18 44                 | 19 09                  | 23 02         | 23 55             | 24 51 | 00 51 |       |       |       |               |              |     |     |     |
| 0                                          | 18 09 | 18 31                 | 18 56                  | 23 15         | 24 11             | 00 11 | 01 08 |       |       |       |               |              |     |     |     |
| S 10                                       | 17 57 | 18 19                 | 18 45                  | 23 28         | 24 27             | 00 27 | 01 26 |       |       |       |               |              |     |     |     |
| 20                                         | 17 45 | 18 08                 | 18 34                  | 23 42         | 24 44             | 00 44 | 01 45 |       |       |       |               |              |     |     |     |
| 30                                         | 17 30 | 17 55                 | 18 24                  | 23 58         | 25 03             | 01 03 | 02 06 |       |       |       |               |              |     |     |     |
| 35                                         | 17 22 | 17 48                 | 18 19                  | 24 07         | 00 07             | 01 15 | 02 19 |       |       |       |               |              |     |     |     |
| 40                                         | 17 12 | 17 41                 | 18 13                  | 24 18         | 00 18             | 01 28 | 02 34 |       |       |       |               |              |     |     |     |
| 45                                         | 17 01 | 17 33                 | 18 08                  | 24 31         | 00 31             | 01 43 | 02 51 |       |       |       |               |              |     |     |     |
| S 50                                       | 16 48 | 17 23                 | 18 02                  | 24 46         | 00 46             | 02 03 | 03 13 |       |       |       |               |              |     |     |     |
| 52                                         | 16 42 | 17 18                 | 17 59                  | 24 53         | 00 53             | 02 12 | 03 23 |       |       |       |               |              |     |     |     |
| 54                                         | 16 35 | 17 14                 | 17 56                  | 25 01         | 01 01             | 02 22 | 03 35 |       |       |       |               |              |     |     |     |
| 56                                         | 16 27 | 17 08                 | 17 53                  | 25 10         | 01 10             | 02 33 | 03 48 |       |       |       |               |              |     |     |     |
| 58                                         | 16 19 | 17 03                 | 17 50                  | 25 21         | 01 21             | 02 47 | 04 03 |       |       |       |               |              |     |     |     |
| S 60                                       | 16 09 | 16 56                 | 17 47                  | 25 32         | 01 32             | 03 02 | 04 22 |       |       |       |               |              |     |     |     |
| D<br>O<br>M<br>I<br>N<br>G<br>O            | SOL   |                       |                        | LUA           |                   |       |       |       |       |       |               |              |     |     |     |
|                                            | Dia   | ET<br>00 <sup>h</sup> | (-)<br>12 <sup>h</sup> | Pass<br>Merid | Pass Merid<br>Sup | Idade | Fase  |       |       |       |               |              |     |     |     |
| d                                          | m s   | m s                   | h m                    | h m           | h m               | d %   |       |       |       |       |               |              |     |     |     |
| 5                                          | 06 06 | 06 03                 | 12 06                  | 17 02         | 04 35             | 06 39 |       |       |       |       |               |              |     |     |     |
| 6                                          | 06 00 | 05 56                 | 12 06                  | 17 58         | 05 30             | 07 51 |       |       |       |       |               |              |     |     |     |
| 7                                          | 05 53 | 05 49                 | 12 06                  | 18 54         | 06 26             | 08 62 |       |       |       |       |               |              |     |     |     |
| SD                                         | 15.8  | d 0.7                 | SD                     | 16.2          | 16.1              | 16.0  |       |       |       |       |               |              |     |     |     |



## ANEXO III

## CONVERSÃO DE ARCO EM TEMPO

| 0°-59° |   |    | 60°-119° |   |    | 120°-179° |    |    | 180°-239° |    |    | 240°-299° |    |    | 300°-359° |    |    | 0°-00 | 0°-25 | 0°-50 | 0°-75 |
|--------|---|----|----------|---|----|-----------|----|----|-----------|----|----|-----------|----|----|-----------|----|----|-------|-------|-------|-------|
| a      | h | m  | o        | h | m  | o         | h  | m  | o         | h  | m  | o         | h  | m  | o         | h  | m  | m     | s     | m     | s     |
| 0      | 0 | 00 | 60       | 4 | 00 | 120       | 8  | 00 | 180       | 12 | 00 | 240       | 16 | 00 | 300       | 20 | 00 | 0     | 00    | 0     | 03    |
| 1      | 0 | 04 | 61       | 4 | 04 | 121       | 8  | 04 | 181       | 12 | 04 | 241       | 16 | 04 | 301       | 20 | 04 | 1     | 04    | 0     | 07    |
| 2      | 0 | 08 | 62       | 4 | 08 | 122       | 8  | 08 | 182       | 12 | 08 | 242       | 16 | 08 | 302       | 20 | 08 | 2     | 08    | 0     | 11    |
| 3      | 0 | 12 | 63       | 4 | 12 | 123       | 8  | 12 | 183       | 12 | 12 | 243       | 16 | 12 | 303       | 20 | 12 | 3     | 12    | 0     | 15    |
| 4      | 0 | 16 | 64       | 4 | 16 | 124       | 8  | 16 | 184       | 12 | 16 | 244       | 16 | 16 | 304       | 20 | 16 | 4     | 16    | 0     | 19    |
| 5      | 0 | 20 | 65       | 4 | 20 | 125       | 8  | 20 | 185       | 12 | 20 | 245       | 16 | 20 | 305       | 20 | 20 | 5     | 20    | 0     | 23    |
| 6      | 0 | 24 | 66       | 4 | 24 | 126       | 8  | 24 | 186       | 12 | 24 | 246       | 16 | 24 | 306       | 20 | 24 | 6     | 24    | 0     | 27    |
| 7      | 0 | 28 | 67       | 4 | 28 | 127       | 8  | 28 | 187       | 12 | 28 | 247       | 16 | 28 | 307       | 20 | 28 | 7     | 28    | 0     | 31    |
| 8      | 0 | 32 | 68       | 4 | 32 | 128       | 8  | 32 | 188       | 12 | 32 | 248       | 16 | 32 | 308       | 20 | 32 | 8     | 32    | 0     | 35    |
| 9      | 0 | 36 | 69       | 4 | 36 | 129       | 8  | 36 | 189       | 12 | 36 | 249       | 16 | 36 | 309       | 20 | 36 | 9     | 36    | 0     | 39    |
| 10     | 0 | 40 | 70       | 4 | 40 | 130       | 8  | 40 | 190       | 12 | 40 | 250       | 16 | 40 | 310       | 20 | 40 | 10    | 40    | 0     | 43    |
| 11     | 0 | 44 | 71       | 4 | 44 | 131       | 8  | 44 | 191       | 12 | 44 | 251       | 16 | 44 | 311       | 20 | 44 | 11    | 44    | 0     | 47    |
| 12     | 0 | 48 | 72       | 4 | 48 | 132       | 8  | 48 | 192       | 12 | 48 | 252       | 16 | 48 | 312       | 20 | 48 | 12    | 48    | 0     | 51    |
| 13     | 0 | 52 | 73       | 4 | 52 | 133       | 8  | 52 | 193       | 12 | 52 | 253       | 16 | 52 | 313       | 20 | 52 | 13    | 52    | 0     | 55    |
| 14     | 0 | 56 | 74       | 4 | 56 | 134       | 8  | 56 | 194       | 12 | 56 | 254       | 16 | 56 | 314       | 20 | 56 | 14    | 56    | 0     | 59    |
| 15     | 1 | 00 | 75       | 5 | 00 | 135       | 9  | 00 | 195       | 13 | 00 | 255       | 17 | 00 | 315       | 21 | 00 | 15    | 100   | 1     | 03    |
| 16     | 1 | 04 | 76       | 5 | 04 | 136       | 9  | 04 | 196       | 13 | 04 | 256       | 17 | 04 | 316       | 21 | 04 | 16    | 104   | 1     | 07    |
| 17     | 1 | 08 | 77       | 5 | 08 | 137       | 9  | 08 | 197       | 13 | 08 | 257       | 17 | 08 | 317       | 21 | 08 | 17    | 108   | 1     | 11    |
| 18     | 1 | 12 | 78       | 5 | 12 | 138       | 9  | 12 | 198       | 13 | 12 | 258       | 17 | 12 | 318       | 21 | 12 | 18    | 112   | 1     | 15    |
| 19     | 1 | 16 | 79       | 5 | 16 | 139       | 9  | 16 | 199       | 13 | 16 | 259       | 17 | 16 | 319       | 21 | 16 | 19    | 116   | 1     | 19    |
| 20     | 1 | 20 | 80       | 5 | 20 | 140       | 9  | 20 | 200       | 13 | 20 | 260       | 17 | 20 | 320       | 21 | 20 | 20    | 120   | 1     | 23    |
| 21     | 1 | 24 | 81       | 5 | 24 | 141       | 9  | 24 | 201       | 13 | 24 | 261       | 17 | 24 | 321       | 21 | 24 | 21    | 124   | 1     | 27    |
| 22     | 1 | 28 | 82       | 5 | 28 | 142       | 9  | 28 | 202       | 13 | 28 | 262       | 17 | 28 | 322       | 21 | 28 | 22    | 128   | 1     | 31    |
| 23     | 1 | 32 | 83       | 5 | 32 | 143       | 9  | 32 | 203       | 13 | 32 | 263       | 17 | 32 | 323       | 21 | 32 | 23    | 132   | 1     | 35    |
| 24     | 1 | 36 | 84       | 5 | 36 | 144       | 9  | 36 | 204       | 13 | 36 | 264       | 17 | 36 | 324       | 21 | 36 | 24    | 136   | 1     | 39    |
| 25     | 1 | 40 | 85       | 5 | 40 | 145       | 9  | 40 | 205       | 13 | 40 | 265       | 17 | 40 | 325       | 21 | 40 | 25    | 140   | 1     | 43    |
| 26     | 1 | 44 | 86       | 5 | 44 | 146       | 9  | 44 | 206       | 13 | 44 | 266       | 17 | 44 | 326       | 21 | 44 | 26    | 144   | 1     | 47    |
| 27     | 1 | 48 | 87       | 5 | 48 | 147       | 9  | 48 | 207       | 13 | 48 | 267       | 17 | 48 | 327       | 21 | 48 | 27    | 148   | 1     | 51    |
| 28     | 1 | 52 | 88       | 5 | 52 | 148       | 9  | 52 | 208       | 13 | 52 | 268       | 17 | 52 | 328       | 21 | 52 | 28    | 152   | 1     | 55    |
| 29     | 1 | 56 | 89       | 5 | 56 | 149       | 9  | 56 | 209       | 13 | 56 | 269       | 17 | 56 | 329       | 21 | 56 | 29    | 156   | 1     | 59    |
| 30     | 2 | 00 | 90       | 6 | 00 | 150       | 10 | 00 | 210       | 14 | 00 | 270       | 18 | 00 | 330       | 22 | 00 | 30    | 200   | 2     | 03    |
| 31     | 2 | 04 | 91       | 6 | 04 | 151       | 10 | 04 | 211       | 14 | 04 | 271       | 18 | 04 | 331       | 22 | 04 | 31    | 204   | 2     | 07    |
| 32     | 2 | 08 | 92       | 6 | 08 | 152       | 10 | 08 | 212       | 14 | 08 | 272       | 18 | 08 | 332       | 22 | 08 | 32    | 208   | 2     | 11    |
| 33     | 2 | 12 | 93       | 6 | 12 | 153       | 10 | 12 | 213       | 14 | 12 | 273       | 18 | 12 | 333       | 22 | 12 | 33    | 212   | 2     | 15    |
| 34     | 2 | 16 | 94       | 6 | 16 | 154       | 10 | 16 | 214       | 14 | 16 | 274       | 18 | 16 | 334       | 22 | 16 | 34    | 216   | 2     | 19    |
| 35     | 2 | 20 | 95       | 6 | 20 | 155       | 10 | 20 | 215       | 14 | 20 | 275       | 18 | 20 | 335       | 22 | 20 | 35    | 220   | 2     | 23    |
| 36     | 2 | 24 | 96       | 6 | 24 | 156       | 10 | 24 | 216       | 14 | 24 | 276       | 18 | 24 | 336       | 22 | 24 | 36    | 224   | 2     | 27    |
| 37     | 2 | 28 | 97       | 6 | 28 | 157       | 10 | 28 | 217       | 14 | 28 | 277       | 18 | 28 | 337       | 22 | 28 | 37    | 228   | 2     | 31    |
| 38     | 2 | 32 | 98       | 6 | 32 | 158       | 10 | 32 | 218       | 14 | 32 | 278       | 18 | 32 | 338       | 22 | 32 | 38    | 232   | 2     | 35    |
| 39     | 2 | 36 | 99       | 6 | 36 | 159       | 10 | 36 | 219       | 14 | 36 | 279       | 18 | 36 | 339       | 22 | 36 | 39    | 236   | 2     | 39    |
| 40     | 2 | 40 | 100      | 6 | 40 | 160       | 10 | 40 | 220       | 14 | 40 | 280       | 18 | 40 | 340       | 22 | 40 | 40    | 240   | 2     | 43    |
| 41     | 2 | 44 | 101      | 6 | 44 | 161       | 10 | 44 | 221       | 14 | 44 | 281       | 18 | 44 | 341       | 22 | 44 | 41    | 244   | 2     | 47    |
| 42     | 2 | 48 | 102      | 6 | 48 | 162       | 10 | 48 | 222       | 14 | 48 | 282       | 18 | 48 | 342       | 22 | 48 | 42    | 248   | 2     | 51    |
| 43     | 2 | 52 | 103      | 6 | 52 | 163       | 10 | 52 | 223       | 14 | 52 | 283       | 18 | 52 | 343       | 22 | 52 | 43    | 252   | 2     | 55    |
| 44     | 2 | 56 | 104      | 6 | 56 | 164       | 10 | 56 | 224       | 14 | 56 | 284       | 18 | 56 | 344       | 22 | 56 | 44    | 256   | 2     | 59    |
| 45     | 3 | 00 | 105      | 7 | 00 | 165       | 11 | 00 | 225       | 15 | 00 | 285       | 19 | 00 | 345       | 23 | 00 | 45    | 300   | 3     | 03    |
| 46     | 3 | 04 | 106      | 7 | 04 | 166       | 11 | 04 | 226       | 15 | 04 | 286       | 19 | 04 | 346       | 23 | 04 | 46    | 304   | 3     | 07    |
| 47     | 3 | 08 | 107      | 7 | 08 | 167       | 11 | 08 | 227       | 15 | 08 | 287       | 19 | 08 | 347       | 23 | 08 | 47    | 308   | 3     | 11    |
| 48     | 3 | 12 | 108      | 7 | 12 | 168       | 11 | 12 | 228       | 15 | 12 | 288       | 19 | 12 | 348       | 23 | 12 | 48    | 312   | 3     | 15    |
| 49     | 3 | 16 | 109      | 7 | 16 | 169       | 11 | 16 | 229       | 15 | 16 | 289       | 19 | 16 | 349       | 23 | 16 | 49    | 316   | 3     | 19    |
| 50     | 3 | 20 | 110      | 7 | 20 | 170       | 11 | 20 | 230       | 15 | 20 | 290       | 19 | 20 | 350       | 23 | 20 | 50    | 320   | 3     | 23    |
| 51     | 3 | 24 | 111      | 7 | 24 | 171       | 11 | 24 | 231       | 15 | 24 | 291       | 19 | 24 | 351       | 23 | 24 | 51    | 324   | 3     | 27    |
| 52     | 3 | 28 | 112      | 7 | 28 | 172       | 11 | 28 | 232       | 15 | 28 | 292       | 19 | 28 | 352       | 23 | 28 | 52    | 328   | 3     | 31    |
| 53     | 3 | 32 | 113      | 7 | 32 | 173       | 11 | 32 | 233       | 15 | 32 | 293       | 19 | 32 | 353       | 23 | 32 | 53    | 332   | 3     | 35    |
| 54     | 3 | 36 | 114      | 7 | 36 | 174       | 11 | 36 | 234       | 15 | 36 | 294       | 19 | 36 | 354       | 23 | 36 | 54    | 336   | 3     | 39    |
| 55     | 3 | 40 | 115      | 7 | 40 | 175       | 11 | 40 | 235       | 15 | 40 | 295       | 19 | 40 | 355       | 23 | 40 | 55    | 340   | 3     | 43    |
| 56     | 3 | 44 | 116      | 7 | 44 | 176       | 11 | 44 | 236       | 15 | 44 | 296       | 19 | 44 | 356       | 23 | 44 | 56    | 344   | 3     | 47    |
| 57     | 3 | 48 | 117      | 7 | 48 | 177       | 11 | 48 | 237       | 15 | 48 | 297       | 19 | 48 | 357       | 23 | 48 | 57    | 348   | 3     | 51    |
| 58     | 3 | 52 | 118      | 7 | 52 | 178       | 11 | 52 | 238       | 15 | 52 | 298       | 19 | 52 | 358       | 23 | 52 | 58    | 352   | 3     | 55    |
| 59     | 3 | 56 | 119      | 7 | 56 | 179       | 11 | 56 | 239       | 15 | 56 | 299       | 19 | 56 | 359       | 23 | 56 | 59    | 356   | 3     | 59    |

A tábua acima destina-se à conversão de arco em tempo; sua principal aplicação nesse Almanaque é a conversão da longitude, cujo valor em horas, minutos e segundos é utilizado na fórmula que relaciona a HML com a TU:  $TU = HML + \lambda$ , sendo  $\lambda$  positivo para longitude W e negativo para longitude E.